

Raw features:

open, high, low, close, volume, timestamp

Engineered features:

Close-to-close return (captures momentum, removes non-stationarity, essential for neural networks and tree models)

$$CCR_t = \frac{close_t - close_{t-1}}{close_{t-1}}$$

Log Return (more statistically stable, better for LSTM/Transformers)

$$cr_t = \ln\left(\frac{close_t}{close_{t-1}}\right)$$

Open-to-close return (measures candle strength, detects bullish/bearish conviction)

$$OCR_t = \frac{close_t - open_t}{open_t}$$

High-low range (normalized candle volatility; strong volatility signal, useful for breakout prediction)

$$HLR_t = \frac{high_t - low_t}{close_t}$$

Cumulative Return (long-term trend detection, useful for regime classification)

$$CR_t = \prod_{i=1}^t (1 + cr_i) - 1$$

Simple Moving Average (trend direction, support/resistance, detects bull/bear phases)

$$SMA_n = \frac{1}{n} \sum_{i=0}^{n-1} close_{t-i}, \quad \text{common windows: 5, 10, 20, 50, 200}$$

Exponential Moving Average (reacts faster than SMA, important for short-term trading)

$$EMA_t = \alpha \cdot close_t + (1 - \alpha) \cdot EMA_{t-1}$$

Price Distance from MA (distance from equilibrium; detects overbought/oversold states, very predictive in crypto)

$$D_t = \frac{close_t - SMA_n}{SMA_n}$$

SMA cross feature (encodes trend regime)

$$SMA_{20} - SMA_{50}$$

Relative Strength Index (one of the strongest crypto indicators, detects momentum exhaustion, strong predictor for reversals)

$$RSI = 100 - \frac{100}{1 + RS}$$

MACD (measures trend acceleration)

$$MACD = EMA_{12} - EMA_{26}$$

Momentum (captures directional strength)

$$Mm_t = close_t - close_{t-n}$$

Rate of Change (measures speed of price movement, good for breakout detection)

$$ROC = \frac{close_t - close_{t-n}}{close_{t-n}} \times 100$$

Rolling Standard Deviation (built From returns; captures uncertainty, essential for regime detection)

$$Volatility_t = \sqrt{\frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})^2}$$

Average True Range (ATR) (excellent volatility estimator)

$$TR_t = \max(high_t - low_t, |high_t - close_{t-1}|, |low_t - close_{t-1}|)$$

$$ATR = EMA(TR)$$

Bollinger Bands Width (built from SMA and rolling std; detects compression before breakouts)

$$BBW = \frac{UpperBand - LowerBand}{SMA}$$

Volume Change (Detects accumulation/distribution)

$$VC_t = \frac{volume_t - volume_{t-1}}{volume_{t-1}}$$

Volume Moving Average (identifies abnormal volume spikes)

$$VSMA_n = \frac{1}{n} \sum_{i=0}^{n-1} volume_{t-i}$$

Relative Volume (Strong breakout confirmation feature)

$$RV_t = \frac{volume_t}{VSMA_n}$$

Price-Volume Trend (PVT) (Combines trend + volume)

$$PVT_t = PVT_{t-1} + volume_t \times \frac{close_t - close_{t-1}}{close_{t-1}}$$

Candle Body Size (Measures market conviction)

$$Body_t = |close_t - open_t|$$

Upper Wick (selling pressure)

$$UpperWick_t = high_t - \max(open_t, close_t)$$

Lower Wick (buying pressure)

$$LowerWick_t = \min(open_t, close_t) - low_t$$

Candle Direction (Useful for classification)

$$Direction_t = \begin{cases} 1 & close_t > open_t \\ 0 & close_t \leq open_t \end{cases}$$

Body-to-Range Ratio (Detects indecision candles)

$$BodyRatio_t = \frac{|close_t - open_t|}{high_t - low_t}$$

Lagged Returns (captures autocorrelation): return(t-1), return(t-2), return(t-24), ...

Lagged Volume (delayed market reaction): volume(t-1), volume(t-5), ...

Lagged Volatility (volatility clustering)

Hour of Day (captures session behavior, US/EU/Asia market effects)

Day of Week (weekend anomalies in BTC)

Month (seasonality)

Cyclical Time Encoding (better representation for ML models)

For hour:

$$HourSin = \sin\left(2\pi \frac{Hour}{24}\right)$$

$$HourCos = \cos\left(2\pi \frac{Hour}{24}\right)$$

Rolling Skewness (asymmetry of returns; detects unstable market phases)

$$Skew_t = \frac{\frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})^3}{\left(\sqrt{\frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})^2}\right)^3}$$

where r_i = returns, $\bar{r}$ = mean return, n = rolling window

Rolling Kurtosis (fat-tail behavior; detects crash-like conditions)

$$Kurtosis_t = \frac{\frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})^4}{\left(\frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})^2\right)^2}$$

Z-Score (measures deviation from normal behavior)

$$Z_t = \frac{close_t - SMA_n}{Std_n}$$

Return × Volume (high-confidence move detection)

$$RetVol = Return_t \times RelVolume_t$$

Volatility × Momentum (detects explosive trend continuation)

$$VolMom = Volatility_t \times Momentum_t$$